(39) Ex:-3:- [B.Q:-4]

A fruit vendor professes to sell fruits @ cost price, but uses false weights & gains 40%. What weight does he substitutes instead of kilogram?

Ans:- SP = CP ⇒ No profit in terms of money.

then $P\% = \frac{\text{Error}}{\text{Original}} \times 100 \Rightarrow 40 = \frac{(1000-x)}{x} \times 100$

$\Rightarrow 40x = 100000 - 100x \Rightarrow 140x = 100000$

$\Rightarrow x = \frac{100000}{140} = 714\left(\frac{2}{7}\right)$ grams

Ex:-4:- [B.Q:-6]:- Rahim went shopping to purchase mobile, shop keeper asked him to pay 18% tax if he wants a bill. If not he can get 7% discount on the actual price of the mobile. Rahim decided to purchase w/o bill and paid 4650/-. Then how much money rahim saved?

Ans:- Lets Assume Mobile Price = 100%.

with tax SP = 100% + 18% = 118%

w/o tax SP = 100% - 7% = 93%

$93\% = 4650 \Rightarrow 1\% = \frac{4650}{93} = 50$

Saved money = 118% - 93% = 25%.

= 25% = 25 × 50 = 1250

Ex:-5:- [B.Q:-8]:- 2 articles purchased @ 8000. 1st @ 40% P, 2nd @ 40% Loss. If the $SP_1 = SP_2$. C.P & cheaper one is?

Ans:- $CP_1 + CP_2 = 8000$ ---① $1.4CP_1 = 0.6CP_2 \Rightarrow$

$CP_2 = \frac{7}{3}CP_1$ ---②

② in ①:-

$CP_1 + \frac{7}{3}CP_1 = 8000 \Rightarrow 10CP_1 = 24000 \Rightarrow$

$CP_1 = 2400$

Ex:-6:-[B.Q:-10]:- A & B are dealers of Honda Bikes. Bike MRP = 28k. 'A' gives a discount of 10% on whole, while 'B' gives a discount of 12% on the first 20k & 8% on the rest 8k. What is diff b/w their S.P's?

Ans:- 'A' SP = 28000 − 10% of 28000 = 28000 − 2800
= 25200

'B' SP = [20000 − 12% of 20k] + [8000 − 8% of 8000]
= [20000 − 2400] + [8000 − 640]
= 17600 + 7360 = 24960

Diff b/w A & B SP's = 25200 − 24960 = 240/-

Ex:-7:-[B.Q:-12]:- Rahul purchased an article @ 8400/- & sold @ loss of 5%. From this money he purchased another article & sold @ 5% gain. Overall gain (or) loss %?

Ans:-
8400
−420 −5% of 8400
7980
+399 +5% of 7980
8379

Net Loss = 8400 − 8379 = 21/-

(or) Net P/L% = $\left[-5+5-\frac{5\times5}{100}\right]$%
= $\frac{-25}{100}$% = 0.25% Loss

Loss = 0.25% of 8400
= 21/-

Ex:-8:-[B.Q:-15:-]:- A reduction of 20% on the sugar price enables a housewife to fetch 6 kgs more for 240/-. What is Original sugar price per kg?

Ans:- Reduced price = 20% of 240/- = 6kgs ⇒ 48/- = 6kgs

20% Reduced Sugar Price Per kgs = 48/6 = 8/- = 0.8 Original

0.8 of Original Sugar kg = 8 ⇒ Original kg = 10/-

Ex:-9:-[B.Q:-16:]:- Suman purchased goods for 900/- & sold $\frac{1}{3}$rd of it @ loss of 12%, then at what gain %. should remaining goods be sold to gain 18% profit on whole(app)?

Ans:- $\frac{1}{3}\times 900\times(1-0.12) + \frac{2}{3}\times 900\times x = (1+\frac{18}{100})900$

(0.12 ↓ Loss; x ↓ P/L to find; 18% profit)

$\Rightarrow \frac{1}{3}\times 900\times 0.88 + \frac{2}{3}\times 900\, x = 900(1.18)$

$264 + 600x = 1062 \Rightarrow x = \frac{1062-264}{600} = \frac{798}{600} = 1.33$

$P\% = (1.33-1)100 = \boxed{33\%\ \text{gain}}$

Ex:-10:-[B.Q:-18:]:- What is the max % discount that a man can offer on his M.R.P, so that he ends up selling at no profit or loss, If he had initially his good up by 35%?

Ans:- x% ↑ed then how much % consumption ↓ed to maintain same expenditure.

$x\% = 35\%$

$\%\downarrow \text{discount} = \frac{35}{(100+35)}\times 100 = \frac{35}{135}\times 100 =$

$= \frac{7}{27}\times 100 = \boxed{26\%}$

Ex:-11:-[B.Q:-19]:- A man purchases few apples @ 3 for 4/- & same quantity @ 4 for 7/-. If he sells all @ 5 for 9/- find his Profit/Loss %?

Ans:- When the question given like this take LCM (3,4,5) as no.of apples in 1st & 2nd cases to do easily.

60 Ap → 3 for 4/- ⇒ 60 Apples = 3×20 = 4×20/- = 80

60 Ap ⇒ 4 for 7/- ⇒ 60A = 4×15 = 7×15 = 105

185

Total Apples S.P = $\frac{120}{5}\times 9 = 216/-$

$P\% = (216-185)/185 \times 100 = \frac{31}{185}\times 100 = \boxed{17\%\ \text{gain}}$